Impedance-Matching vs. Robin-Robin Coupling Conditions in Schwarz Methods

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1 Setting

Consider a solid body $\Omega \subset \mathbb{R}^3$ decomposed into two subdomains Ω_1 and Ω_2 that are integrated independently during a Schwarz iteration. Two geometric configurations are used in Norma.jl:

Non-overlap. The subdomains partition Ω and share an interface $\Gamma = \partial\Omega_1 \cap \partial\Omega_2$ on which coupling is imposed. Let $\boldsymbol{n}_k$ be the outward unit normal to Ω_k on Γ (so $\boldsymbol{n}_1 = -\boldsymbol{n}_2$), and let

$$\boldsymbol{t}_k \;=\; \boldsymbol{\sigma}_k\, \boldsymbol{n}_k$$

denote the traction on Ω_k's side of the interface. Continuity of the coupled solution requires

$$\boldsymbol{u}_1 \;=\; \boldsymbol{u}_2 \qquad \text{(kinematic continuity)}, \tag{1}$$

$$\boldsymbol{t}_1 + \boldsymbol{t}_2 \;=\; \boldsymbol{0} \qquad \text{(traction balance)}. \tag{2}$$

Classical Dirichlet-Neumann (DN) Schwarz picks up (1) on one side and (2) on the other. The *Robin-Robin* (RR) and *impedance* variants each replace both Dirichlet and Neumann conditions with a linear combination of displacement and traction, which offers extra tuning and, in the impedance case, wave absorption.

Overlap. The subdomains cover parts of Ω that properly intersect: $\Omega_1 \cap \Omega_2 \neq \emptyset$. Each Ω_k has a portion of its boundary, $\Gamma_k \subset \Omega_j$ with $j \neq k$, that lies strictly in the interior of the partner. That boundary is not shared with Ω_j, so no mutual traction is available there; on the other hand, the partner provides a well-defined interior state at every node of Γ_k. Classical overlap Schwarz enforces kinematic matching

$$\boldsymbol{u}_k \;=\; \boldsymbol{u}_j \qquad \text{on } \Gamma_k,$$

either pointwise (strong interpolation) or in an L^2 sense (weak projection). The *impedance overlap* variant instead leaves the overlap degrees of freedom free and drives $\boldsymbol{u}_k$ towards $\boldsymbol{u}_j$ through an impedance-matching penalty, as detailed in Section 4.

2 Robin-Robin (RR) Non-Overlap Schwarz

2.1 Continuous formulation

On each side of the interface, the user specifies a Robin parameter $\alpha > 0$ and imposes

$$\boldsymbol{t}_k + \alpha\, \boldsymbol{u}_k \;=\; \boldsymbol{g}_k \qquad \text{on } \Gamma, \quad k = 1, 2, \tag{3}$$

with the forcing $\boldsymbol{g}_k$ set from the partner subdomain's state at the previous Schwarz iterate:

$$\boldsymbol{g}_k^{(n+1)} \;=\; -\,\boldsymbol{t}_j^{(n)} + \alpha\,\boldsymbol{u}_j^{(n)} \qquad j \neq k. \tag{4}$$

When the Schwarz iteration converges, $\boldsymbol{u}_k = \boldsymbol{u}_j$ and $\boldsymbol{t}_k = -\boldsymbol{t}_j$ so that (3) reproduces (1) and (2) simultaneously. The combined "elastic-spring" interface smooths out the DN mismatch that drives DN Schwarz towards a fixed point.

2.2 Discrete implementation

Let $\boldsymbol{W} = \int_\Gamma \boldsymbol{N}\,\boldsymbol{N}^\top \,\mathrm{d}\Gamma$ be the surface mass (square-projector) matrix and let $\boldsymbol{P}$ be the Dirichlet projector mapping the partner's interface trace to this subdomain's interface nodes. The RR boundary condition contributes to the solid subproblem on Ω_k as follows:

Self stiffness term. From (3), the quantity $\alpha\,\boldsymbol{u}_k$ acts as an elastic boundary stiffness which is assembled into the global tangent:

$$\boldsymbol{K}_{\mathrm{rs}} \;=\; \alpha\,\boldsymbol{W} \qquad \text{(diagonal of } \Gamma\text{)}. \tag{5}$$

Partner contribution (right-hand side). The forcing $\boldsymbol{g}_k$ from (4) is accumulated into the boundary-force vector of Ω_k:

$$\boldsymbol{f}_{\Gamma,k} \;\leftarrow\; \boldsymbol{f}_{\Gamma,k} \;-\; \boldsymbol{t}_j^{(n)}\big|_\Gamma \;+\; \alpha\,\boldsymbol{W}\,\boldsymbol{P}\,\boldsymbol{u}_j^{(n)}. \tag{6}$$

In Norma.jl the partner traction is negated once by `get_dst_force` and the self term $\alpha\,\boldsymbol{W}\,\boldsymbol{u}_k$ is added back via `apply_robin_bcs_internal_force!`.

2.3 Parameter tuning

α has units of stiffness per unit area (Pa/m = N/m^3). It trades off traction and displacement error:

- $\alpha \to \infty$: the Robin condition approaches a pure Dirichlet constraint; no traction feedback from the partner.
- $\alpha \to 0$: the Robin condition approaches a pure Neumann (traction) constraint; no displacement feedback.
- An optimal α minimizes the Schwarz convergence rate. Closed-form expressions have been studied in the optimized-Schwarz literature but depend on details of the discretization and problem. In practice α is tuned empirically; the scaling $\alpha \sim E/h$ (Young's modulus over a representative element size) has been effective on the problems exercised here.

2.4 Limitations for dynamic problems

For dynamic (wave-propagation) problems, (3) behaves as a purely elastic interface. An incoming wave from Ω_j is partly reflected back into Ω_j and partly transmitted into Ω_k, with reflection coefficient determined by the interface impedance mismatch. Because α has no physical correspondence to the material's wave impedance, reflections in general are non-zero and can drive energy growth when the two subdomains use different time integrators (e.g. an implicit/explicit pairing across the interface).

3 Impedance Non-Overlap Schwarz

3.1 Motivation: non-reflecting boundary in 1D

Consider a semi-infinite elastic bar with density ρ and P-wave modulus $M = \lambda + 2\mu$, supporting a right-going traveling wave $u(x,t) = f(x - c_p t)$ with $c_p = \sqrt{M/\rho}$. Differentiating in x and t gives the kinematic identity $\dot{u} = u_t = -c_p\, u_x$. Combined with 1D Hooke's law $\sigma = M\, u_x$ this yields

$$\sigma \;=\; M\, u_x \;=\; -\tfrac{M}{c_p}\,\dot{u} \;=\; -\,\rho\, c_p\, \dot{u}. \tag{7}$$

Define the *characteristic acoustic impedance*

$$Z \;=\; \rho\, c_p \;=\; \sqrt{\rho\,(\lambda + 2\mu)}. \tag{8}$$

Then for a purely outgoing (right-going) wave,

$$\sigma + Z\,\dot{u} \;=\; 0, \tag{9}$$

which is the Lysmer-Kuhlemeyer non-reflecting boundary condition. Imposing (9) on the interface prevents spurious reflections of outgoing waves and is the key ingredient in the impedance variant.

3.2 Continuous formulation

Norma.jl's impedance Schwarz BC augments the Robin condition with the impedance term from (9):

$$\boldsymbol{t}_k + Z\,\dot{\boldsymbol{u}}_k + \alpha\,\boldsymbol{u}_k \;=\; \boldsymbol{g}_k \qquad \text{on } \Gamma. \tag{10}$$

The forcing is built from the partner state:

$$\boldsymbol{g}_k^{(n+1)} \;=\; -\,\boldsymbol{t}_j^{(n)} \;+\; Z\,\dot{\boldsymbol{u}}_j^{(n)} \;+\; \alpha\,\boldsymbol{u}_j^{(n)}. \tag{11}$$

At convergence, $\boldsymbol{u}_k = \boldsymbol{u}_j$, $\dot{\boldsymbol{u}}_k = \dot{\boldsymbol{u}}_j$, and $\boldsymbol{t}_k = -\boldsymbol{t}_j$, so the impedance and displacement-penalty terms cancel and (10) again reproduces continuity and traction balance.

Setting $\alpha = 0$ yields the pure impedance condition; the displacement-penalty term $\alpha\,\boldsymbol{u}$ is retained as an option to regularize the quasi-static limit, where $\dot{\boldsymbol{u}} \to \boldsymbol{0}$ and (10) would otherwise reduce to the Neumann condition only.

3.3 Discrete implementation

For a Newmark integrator with parameters β, γ and time step Δt, the velocity is a linear function of displacement at the new step,

$$\dot{\boldsymbol{u}}_k^{(n+1)} \;=\; \frac{\gamma}{\beta\,\Delta t}\,\boldsymbol{u}_k^{(n+1)} \;+\; (\text{terms known at } t_n). \tag{12}$$

Let $c_v \;=\; \gamma/(\beta\,\Delta t)$. The self term $Z\,\dot{\boldsymbol{u}}_k + \alpha\,\boldsymbol{u}_k$ therefore produces the tangent contribution

$$\boldsymbol{K}_{\text{is}} \;=\; \big(Z\,c_v + \alpha\big)\,\boldsymbol{W}, \tag{13}$$

which is assembled into the global Hessian at each Newton iteration. In the quasi-static limit Norma.jl sets $c_v = 0$, so (13) reduces to $\alpha\,\boldsymbol{W}$ and the impedance BC collapses onto RR.

The self internal-force contribution at the current state is

$$\boldsymbol{f}_{\text{is}} \;=\; \boldsymbol{W}\,\big(\,Z\,\dot{\boldsymbol{u}}_k \;+\; \alpha\,\boldsymbol{u}_k\,\big), \tag{14}$$

added by `build_impedance_schwarz_force` (`schwarz.jl`).

The partner contribution is accumulated into the boundary-force vector:

$$\boldsymbol{f}_{\Gamma,k} \;\leftarrow\; \boldsymbol{f}_{\Gamma,k} \;-\; \boldsymbol{t}_j^{(n)}\big|_{\Gamma} \;+\; Z\,\boldsymbol{W}\,\boldsymbol{P}\,\dot{\boldsymbol{u}}_j^{(n)} \;+\; \alpha\,\boldsymbol{W}\,\boldsymbol{P}\,\boldsymbol{u}_j^{(n)}, \tag{15}$$

as implemented in `apply_bc_detail` for `SolidMechanicsImpedanceSchwarzBoundaryCondition` (`schwarz.jl`).

3.4 Derivation of Z from material properties

In Norma.jl the impedance is computed per-domain from the material parameters:

$$Z \;=\; \sqrt{\rho\,(\lambda + 2\mu)} \;=\; \rho\, c_p, \qquad \text{with } c_p = \sqrt{(\lambda+2\mu)/\rho}. \tag{16}$$

No user tuning is required for the impedance term; only the optional displacement-penalty α is exposed via the `robin parameter` field.

4 Overlap Schwarz

On an overlap boundary Γ_k, the partner field $\boldsymbol{u}_j$ is available but the partner *traction* is not, because Γ_k is not a shared interface: it is a cut through the interior of Ω_j. Only those coupling conditions that do not require a partner traction can be imposed on an overlap boundary. Norma.jl provides two such families.

4.1 Classical overlap (Dirichlet coupling)

The canonical overlap Schwarz BC replaces the free boundary of Ω_k on Γ_k with an imposed kinematic trace drawn from the partner,

$$\boldsymbol{u}_k \;=\; \boldsymbol{u}_j, \qquad \dot{\boldsymbol{u}}_k \;=\; \dot{\boldsymbol{u}}_j, \qquad \ddot{\boldsymbol{u}}_k \;=\; \ddot{\boldsymbol{u}}_j \qquad \text{on } \Gamma_k. \tag{17}$$

The trace is constructed in one of two ways:

Strong interpolation. For each overlap node $\boldsymbol{x}_i \in \Gamma_k$, find the partner element containing $\boldsymbol{x}_i$ in its reference configuration, evaluate the partner shape functions $\boldsymbol{N}_j(\boldsymbol{x}_i)$, and assign

$$\boldsymbol{u}_k(\boldsymbol{x}_i) \;=\; \sum_a \boldsymbol{N}_j^a(\boldsymbol{x}_i)\,\boldsymbol{u}_j^a.$$

The corresponding degrees of freedom on Ω_k are then frozen.

Weak (L^2) projection. A Dirichlet projector $\boldsymbol{P}$ is precomputed by solving the boundary L^2 problem $\boldsymbol{W}_k\,\hat{\boldsymbol{u}}_k = \boldsymbol{C}_{kj}\,\boldsymbol{u}_j$, where $\boldsymbol{W}_k$ is the mass matrix of Γ_k and $\boldsymbol{C}_{kj}$ is the cross-mass coupling the partner's shape functions to the overlap boundary. Overlap-node displacements are then assigned $\boldsymbol{u}_k = \boldsymbol{P}\,\boldsymbol{u}_j$.

Both variants are implemented by `SolidMechanicsOverlapSchwarzBoundaryCondition` and selected via the `use_weak` flag (`coupling_strong_dbc`, `coupling_weak_overlap_dbc` in `schwarz.jl`). Because (17) constrains the overlap DOFs, classical overlap adds no interface tangent term: the constrained rows and columns are eliminated from the system.

4.2 Impedance overlap

The impedance overlap BC, by contrast, leaves the overlap DOFs *free* and couples Ω_k to the partner through a full zeroth-order optimized-Schwarz (Robin) transmission condition,

$$\boldsymbol{t}_k \;-\; \boldsymbol{t}_j \;+\; \boldsymbol{Z}\left(\dot{\boldsymbol{u}}_k - \dot{\boldsymbol{u}}_j\right) \;+\; \alpha\left(\boldsymbol{u}_k - \boldsymbol{u}_j\right) \;=\; \boldsymbol{0} \qquad \text{on } \Gamma_k, \tag{18}$$

where $\boldsymbol{t}_j = \boldsymbol{\sigma}_j \boldsymbol{n}_k$ is the partner's traction evaluated on Γ_k with Ω_k's outward normal, and the impedance is the P/S-split tensor

$$\boldsymbol{Z} \;=\; Z_p\, \boldsymbol{n}_k \otimes \boldsymbol{n}_k \;+\; Z_s\left(\boldsymbol{I} - \boldsymbol{n}_k \otimes \boldsymbol{n}_k\right), \qquad Z_p = \sqrt{\rho(\lambda + 2\mu)}, \quad Z_s = \sqrt{\rho\mu}, \tag{19}$$

with ρ, λ, μ taken from the *partner's* material (optimized-Schwarz cross-scaling: each side's optimal transmission operator approximates the neighbor's Dirichlet-to-Neumann map, so at a bimaterial interface each subdomain carries the other's characteristic impedances).

The partner-traction term in (18) is essential for consistency. The true monodomain solution has continuous kinematics and a generally nonzero traction $\boldsymbol{t}$ on the artificial surface Γ_k; it satisfies (18) identically. Dropping $-\boldsymbol{t}_j$ (as an earlier implementation did) turns the condition into a relative dashpot, $\boldsymbol{t}_k = \boldsymbol{Z}(\dot{\boldsymbol{u}}_j - \dot{\boldsymbol{u}}_k) + \cdots$, which the monodomain solution fails by exactly $\boldsymbol{t}$: the converged coupled solution then carries a spurious interface velocity jump $\approx \boldsymbol{t}/Z$ whose power drain $\sim |\boldsymbol{t}|^2/Z$ acts as permanent interface damping (56% energy loss over 1 ms on a conforming benchmark; see Section 6).

Because Γ_k is an interior surface of the partner mesh, the partner's assembled internal force cannot supply $\boldsymbol{t}_j$ there (interior rows carry the inertia residual, not $\boldsymbol{\sigma n}$). Norma.jl evaluates $\boldsymbol{t}_j$ in one of two ways, selected by the `partner traction` option of the BC (default `auto`).

Variationally consistent traction (node-aligned interfaces). When every node of Γ_k coincides with a partner node (detected automatically from the interpolation weights), the weak partner traction is extracted as a *consistent boundary flux* (Wheeler; Hughes): for each interface node i,

$$\left[\boldsymbol{W}\, \boldsymbol{t}_j\right]_i \;=\; \int_{\Gamma_k} \boldsymbol{t}_j\, \varphi_i \, \mathrm{d}\Gamma \;=\; -\sum_{e \,\in\, \mathcal{E}_i^{\mathrm{ext}}} \left(\boldsymbol{f}_e^{\mathrm{int}} + \boldsymbol{M}_e\, \ddot{\boldsymbol{u}}_e\right)_i, \tag{20}$$

where $\mathcal{E}_i^{\mathrm{ext}}$ is the single layer of partner elements adjacent to Γ_k on its *exterior* side — the side Ω_k's outward normal points toward, i.e. precisely the elements Ω_k is missing from the monodomain assembly — and $\boldsymbol{M}_e$ is the partner's element mass (consistent or lumped, matching the partner's integrator). With this definition the monodomain solution satisfies Ω_k's coupled discrete equations *identically*: the transmission condition transmits all discrete content, including the mesh-scale part that nodal stress recovery misrepresents. Since the interface is node-aligned, the surface operators of the two meshes coincide on Γ_k and (20) enters the right-hand side directly, with no mass-matrix inversion and no recovery solve. Implemented by `build_flux_traction_patch!` (setup; rebuilt on BC swaps) and `impedance_flux_partner_traction` (`schwarz.jl`).

Recovered-stress traction (fallback). On non-conforming interfaces $\boldsymbol{t}_j$ is obtained by interpolating the partner's nodal-recovered stress with the same shape-function data used for $\dot{\boldsymbol{u}}_j$ and $\boldsymbol{u}_j$, and contracting with Ω_k's outward surface normals. *Consistent* (L^2-projected, Cholesky-factorized) nodal recovery is force-enabled for any model that carries this BC type; lumped recovery's $O(h)$ nodal-stress error feeds directly into the transmission condition and was measured to dominate the residual energy error (13.6% vs. 1.5% with consistent recovery; Section 6). Even consistent recovery cannot represent the traction carried by mesh-scale field content, which leaves a first-order-in-h interface dissipation that the consistent-flux extraction removes (Section 6).

The discrete assembly:

- **Self tangent.** $3{\times}3$ nodal blocks $\boldsymbol{K}_{\mathrm{is},ij} = W_{ij}\left(c_v \boldsymbol{Z}(\boldsymbol{n}_j) + \alpha \boldsymbol{I}\right)$, with $c_v = \gamma/(\beta \Delta t)$ under Newmark and the tensor evaluated at the source node's normal, assembled by `build_impedance_overlap_schwarz_stiffness`.

- **Self internal force.** $\boldsymbol{f}_{\mathrm{is}} = \boldsymbol{W}\left(\boldsymbol{Z}\,\dot{\boldsymbol{u}}_k + \alpha\,\boldsymbol{u}_k\right)$, with $\boldsymbol{Z}$ applied nodally before the surface weighting, assembled by `build_impedance_schwarz_force`.

- **Partner right-hand side.** $\boldsymbol{f}_\Gamma \leftarrow \boldsymbol{f}_\Gamma + \boldsymbol{W}\,\boldsymbol{t}_j + \boldsymbol{W}\left(\boldsymbol{Z}\,\dot{\boldsymbol{u}}_j + \alpha\,\boldsymbol{u}_j\right)$, where $\dot{\boldsymbol{u}}_j$ and $\boldsymbol{u}_j$ are obtained by strong interpolation from the partner element containing each overlap node, and $\boldsymbol{W}\boldsymbol{t}_j$ is the consistent-flux weak traction (20) on node-aligned interfaces or $\boldsymbol{W}$ applied to the recovered-stress traction otherwise (`apply_bc_detail` for `SolidMechanicsImpedanceOverlapSchwarzBoundaryCondit`

At the Schwarz fixed point all three jumps in (18) vanish for the monodomain solution, so on conforming meshes the converged coupled solution is the monodomain solution, and the condition needs no parameter tuning: $\boldsymbol{Z}$ is fixed by the (partner) material and α may be zero.

Norma.jl additionally allows a step-dependent multiplier on $\boldsymbol{Z}$ through the `impedance_scale` vector, applied uniformly to the partner term, the self force, and the tangent, so a scaled run rescales the whole operator. With the traction term in place the conserving scale is the physical one (scale = 1); the historical role of the scale as a per-problem tuning knob compensated the missing-traction dissipation and is obsolete.

4.3 Why there is no Robin-Robin overlap variant

An RR condition, $\boldsymbol{t} + \alpha\,\boldsymbol{u} = \boldsymbol{g}$, requires a partner traction to build its right-hand side $\boldsymbol{g} = -\boldsymbol{t}_j + \alpha\,\boldsymbol{u}_j$. On an overlap boundary the partner's assembled internal force cannot supply that traction (the boundary is a cut through the partner's interior), but the recovered-stress interpolation introduced for (18) now provides it. An RR overlap condition is therefore exactly the $\boldsymbol{Z} = \boldsymbol{0}$ special case of (18),

$$\boldsymbol{t}_k - \boldsymbol{t}_j + \alpha\,(\boldsymbol{u}_k - \boldsymbol{u}_j) \;=\; \boldsymbol{0},$$

a traction-consistent displacement penalty with no dynamic absorption. It is subsumed by the impedance overlap BC (set the impedance scale to zero, keep α) and is not exposed as a separate option.

5 Comparison

5.1 Side-by-side formulation

Non-overlap variants.

	RR	Impedance
Boundary condition	$\boldsymbol{t}_k + \alpha\,\boldsymbol{u}_k = \boldsymbol{g}_k$	$\boldsymbol{t}_k + Z\,\dot{\boldsymbol{u}}_k + \alpha\,\boldsymbol{u}_k = \boldsymbol{g}_k$
Self tangent	$\alpha\,\boldsymbol{W}$	$(Z\,c_v + \alpha)\,\boldsymbol{W}$
Self internal force	$\alpha\,\boldsymbol{W}\,\boldsymbol{u}_k$	$\boldsymbol{W}\,\big(Z\,\dot{\boldsymbol{u}}_k + \alpha\,\boldsymbol{u}_k\big)$
Partner RHS	$-\boldsymbol{t}_j + \alpha\,\boldsymbol{W}\,\boldsymbol{P}\,\boldsymbol{u}_j$	$-\boldsymbol{t}_j + Z\,\boldsymbol{W}\,\boldsymbol{P}\,\dot{\boldsymbol{u}}_j + \alpha\,\boldsymbol{W}\,\boldsymbol{P}\,\boldsymbol{u}_j$
Partner traction $\boldsymbol{t}_j$	used	used
Z set by	(not used)	material (auto)
α set by	user	user, optional
Quasi-static limit	same	collapses to RR

Overlap variants.

	Classical (DBC)	Impedance
Boundary condition	$\boldsymbol{u}_k = \boldsymbol{u}_j$ (strong or L^2)	$\boldsymbol{t}_k - \boldsymbol{t}_j + \boldsymbol{Z}\,(\dot{\boldsymbol{u}}_k - \dot{\boldsymbol{u}}_j) + \alpha\,(\boldsymbol{u}_k - \boldsymbol{u}_j) = \boldsymbol{0}$
Self tangent	(DOFs eliminated)	$W_{ij}\,\big(c_v\boldsymbol{Z}(\boldsymbol{n}_j) + \alpha\boldsymbol{I}\big)$ (3×3 blocks)
Self internal force	(none)	$\boldsymbol{W}\,\big(\boldsymbol{Z}\,\dot{\boldsymbol{u}}_k + \alpha\,\boldsymbol{u}_k\big)$
Partner RHS	$\boldsymbol{u}_k \leftarrow \boldsymbol{P}\,\boldsymbol{u}_j$	$\boldsymbol{W}\,\big(\boldsymbol{t}_j + \boldsymbol{Z}\,\dot{\boldsymbol{u}}_j + \alpha\,\boldsymbol{u}_j\big)$
Partner traction $\boldsymbol{t}_j$	(not available)	interpolated nodal-recovered stress $\cdot\,\boldsymbol{n}_k$
$\boldsymbol{Z}$ set by	(not used)	partner material (auto), Z_p/Z_s split, optional scale
α set by	(not used)	user, optional
Quasi-static limit	same	traction-consistent α-penalty ($c_v = 0$)

5.2 Reflection at a non-overlap interface

The reflection analysis below is for the non-overlap geometry, where a single shared interface carries both the self and the partner traction. In the overlap case each subdomain's coupling boundary is embedded in the partner's interior, so the analysis below does not apply directly; Section 4 discusses the overlap behavior separately.

Place the interface at $x = 0$ and the subdomain Ω_k in $x < 0$, so the outward normal of Ω_k at Γ is $+\hat{x}$. Look for harmonic plane-wave solutions of the 1D wave equation: a right-going incident wave u_{inc} propagating out of Ω_k toward Γ, and a left-going reflected wave u_{ref} propagating back into Ω_k,

$$u_{\mathrm{inc}}(x,t) = F\,\mathrm{e}^{\mathrm{i}(\omega t - kx)}, \qquad u_{\mathrm{ref}}(x,t) = R\,\mathrm{e}^{\mathrm{i}(\omega t + kx)}.$$

Here ω is the angular frequency, $k = \omega/c_p$ is the wavenumber from the 1D elastic dispersion relation, and F, R are complex amplitudes. Substituting into $\sigma = M\,u_x$ and using $M\,k = M\,\omega/c_p = \rho c_p^2 \cdot \omega/c_p = Z\,\omega$ gives the interface values

$$u(0,t) = (F+R)\,\mathrm{e}^{\mathrm{i}\omega t}, \quad \sigma(0,t) = \mathrm{i}\,\omega\,Z\,(R-F)\,\mathrm{e}^{\mathrm{i}\omega t}, \quad \dot{u}(0,t) = \mathrm{i}\,\omega\,(F+R)\,\mathrm{e}^{\mathrm{i}\omega t}.$$

RR boundary condition. Imposing $\sigma + \alpha\,u = 0$ at $x = 0$ gives

$$\mathrm{i}\,Z\,\omega\,(R-F) \;+\; \alpha\,(F+R) \;=\; 0,$$

and solving for the reflection coefficient $R_{\mathrm{RR}}(\omega) \equiv R/F$ yields

$$R_{\mathrm{RR}}(\omega) \;=\; \frac{\mathrm{i}\,\omega\,Z - \alpha}{\mathrm{i}\,\omega\,Z + \alpha}. \tag{21}$$

For any real $\alpha > 0$ the numerator and denominator have the same modulus, so $|R_{\mathrm{RR}}(\omega)| \equiv 1$: the elastic interface *fully reflects* the incoming wave with a frequency-dependent phase shift. No wave energy crosses the interface.

Impedance boundary condition. Imposing $\sigma + Z\,\dot{u} + \alpha\,u = 0$ at $x = 0$ gives

$$\mathrm{i}\,Z\,\omega\,(R - F) \;+\; \mathrm{i}\,Z\,\omega\,(F + R) \;+\; \alpha\,(F + R) \;=\; 0,$$

which simplifies to $(2\,\mathrm{i}\,Z\,\omega + \alpha)\,R + \alpha\,F = 0$, so

$$R_{\mathrm{imp}}(\omega) \;=\; -\,\frac{\alpha}{2\,\mathrm{i}\,\omega\,Z + \alpha}. \tag{22}$$

With $\alpha = 0$ the pure impedance condition is satisfied identically by an outgoing wave, and $R_{\mathrm{imp}} \equiv 0$: the wave is absorbed at all frequencies. For $\alpha > 0$, $|R_{\mathrm{imp}}(\omega)| = \alpha/\sqrt{\alpha^2 + 4\,\omega^2\,Z^2}$, which is strictly below one at every frequency and tends to zero as $\alpha \to 0$.

Comparison. The contrast between (21) and (22) is the key difference between the two Schwarz variants. Classical RR conserves energy at the interface ($|R_{\mathrm{RR}}| \equiv 1$), reflecting outgoing waves back into the originating subdomain; impedance Schwarz dissipates outgoing-wave energy into the partner ($|R_{\mathrm{imp}}| < 1$, and $R_{\mathrm{imp}} = 0$ at $\alpha = 0$). This is the mechanism by which impedance Schwarz eliminates the reflection-driven instability that plagues classical RR in time-dependent problems.

5.3 Energy behavior with mixed integrators

With classical RR coupling across an implicit/explicit interface, reflections from the elastic interface can beat against each other and cause non-monotone energy growth over many Schwarz iterations. The impedance variant shunts the outgoing wave energy into the partner and removes that pathway, which stabilizes dynamic simulations that integrate the two subdomains differently (e.g. Newmark + central difference) without requiring a global time step.

5.4 Quasi-static limit

For quasi-static problems ($\dot{\boldsymbol{u}} \to \mathbf{0}$, $c_v = 0$ in (13)), the impedance condition (10) reduces exactly to the Robin condition (3). Norma.jl's quasi-static branch computes $c_v = 0$ explicitly and the impedance pipeline assembles the same $\alpha\,\boldsymbol{W}$ tangent as RR. In particular, an impedance run with $\alpha = 0$ in the quasi-static limit degenerates to a pure Neumann condition and does not couple the subdomains through displacement, which is an important edge case to avoid in static use.

6 Numerical findings (cantilever benchmark)

The statements in this section come from a systematic study of a bent 3D cantilever released from a parabolic initial displacement ($E = 6.895\,\mathrm{GPa}$, $\nu = 0.25$, $\rho = 2768\,\mathrm{kg/m^3}$, bar length $0.254\,\mathrm{m}$, two overlapping subdomains with a 20% overlap, $\Delta t = 10^{-6}\,\mathrm{s}$, $t_{\mathrm{final}} = 10^{-3}\,\mathrm{s}$, Schwarz iterated to convergence at relative tolerance 10^{-8} every step — always 2 iterations). Runs used trapezoidal Newmark (I) and central difference (E) in the four subdomain pairings II, EE, IE, EI, on a non-conforming mesh pair ($h = 8.47$ vs. $6.35\,\mathrm{mm}$) and a conforming, node-aligned pair. Data and scripts reside on `rigel:~/test/Norma/cantilever-energy`.

Energy metric. Summing the two subdomains' energies counts the overlap region twice and pollutes the diagnostic: a provably conservative run shows apparent ±13% excursions as the pulse transits the overlap. All figures quoted here use a partition-of-unity (PU) energy in which elements whose centroid lies in the region covered by both meshes carry weight 1/2, in both $E(t)$ and E_0 (post-processor `pu_energy.jl`; its unweighted sum reproduces Norma's own energy output to at least four digits).

Controls. Converged classical (DBC) overlap Schwarz on *conforming* meshes conserves the PU energy to 0.1% over 1000 steps: at the fixed point the coupled solution is the monodomain solution, exactly as the theory requires. On non-conforming meshes the same scheme is exponentially unstable (energy growth ×27–80 at rate $\sim 7 \times 10^3$ /s; strong and weak L^2 enforcement share the same terminal rate, and a 2:1 *nested* pair is as unstable as the 4:3 non-nested pair), so the instability is caused by the fine-to-coarse information loss of the interface transfer, not by the enforcement or nestedness.

Impedance overlap: the error ladder. On the conforming control (II, PU metric, physical impedance, $\alpha = 0$), the energy error over 1 ms was

$$\underbrace{59\%}_{\text{missing } \boldsymbol{t}_j} \longrightarrow \underbrace{14\%}_{+\,\text{traction term (lumped recovery)}} \longrightarrow \underbrace{1.5\%}_{+\,\text{consistent recovery}} \longrightarrow \underbrace{0.08\%}_{+\,\text{consistent-flux traction}}$$

against the 0.1% exact-DBC reference, which the final rung matches. All corrections are parameter-free.

Residual-dissipation diagnosis. The 1.5–4% residual of the recovered-stress rung was isolated with an interface-work instrumentation (`report_impedance_interface_work`, env-gated) that decomposes the interface power into partner-traction, dashpot, and Robin parts; the dashpot and Robin powers vanish identically for the exact solution, so their integrated work measures the spurious dissipation directly. Three experiments discriminated the candidate mechanisms: (i) sweeping the Schwarz tolerance from 10^{-8} to 10^{-11} leaves E/E_0 *bit-identical* — the Gauss–Seidel iteration reaches machine precision in two passes at every step, ruling out iteration lag; (ii) the loss is invariant under $\Delta t \in \{2, 1, 0.5\} \times 10^{-6}$ s, ruling out the Newmark time level; (iii) the loss halves when h is halved (3.9% → 1.9%): first order in mesh size. The measured velocity jump satisfies $\boldsymbol{Z}(\dot{\boldsymbol{u}}_k - \dot{\boldsymbol{u}}_j) \approx \boldsymbol{t}_j - \boldsymbol{t}_k$ throughout — exactly what the converged transmission condition dictates — so the mechanism is: the recovered-stress traction error on mesh-scale field content is converted by the Robin condition into an interface velocity jump $\sim \delta t/Z$, which the dashpot dissipates at rate $|\delta t|^2/Z$, steadily over the whole run. Replacing the recovered traction with the consistent flux (20) eliminates it: energy loss 3.93% → 0.076% and integrated dashpot work −42.6 J → +0.1 J on the same benchmark (tensor impedance). This also explains why the classical DBC overlap is exact on conforming meshes — nodal injection transmits all discrete content with no recovery step — and closes the gap between the two.

Impedance scale. With the traction term missing, a swept constant multiplier on Z interpolates Neumann-like (over-absorbing) to Dirichlet-like (unstable) behavior, and the best scale on the non-conforming pair was 1.12 — a cancellation of the missing-traction dissipation against the non-conforming transfer injection, hence mesh- and problem-specific. With the traction term in place the optimum is the physical scale 1.0, and the historical per-simulation tuning of the impedance scale (and of large α values near 5×10^{11}–10^{12}) is obsolete: those tuned values optimized the double-counted diagnostic, not the physics (the tuned pre-fix best measures 23.6% in the PU metric vs. 7.1% for the corrected parameter-free condition on the same meshes).

P/S tensor split. The tensor impedance (19) is exact for body waves at normal incidence and is the theoretically correct operator for bimaterial interfaces. With the recovered-stress traction it appeared slightly *detrimental* on this homogeneous, flexure-dominated benchmark (conforming: 4.1% vs. 1.5% scalar), and its apparent advantage on the non-conforming pair (3.9% vs. 4.4%) was a cancellation against transfer injection. The diagnosis above resolves this: the penalty was never a property of the impedance split — the tangential traction (recovery) error, which dominates in flexure, is dissipated at rate $|\delta t|^2/Z_s$, and $Z_s = Z_p/\sqrt{3}$ amplifies it. With the consistent-flux traction the error term is gone and the tensor impedance conserves to 0.076% on the same benchmark. Transmission conditions should be benchmarked on conforming meshes; non-conforming results flatter the scheme.

Open items. (i) Non-conforming interfaces still use the recovered-stress traction and pointwise interpolation; the ∼4% transfer injection remains (a mortar transfer with a flux-consistent pairing, in the spirit of (20), is the candidate fix). (ii) Pairings with an explicit subdomain lose ∼8–10% of the PU energy within the first few steps and oscillate per step thereafter (∼15% overall with consistent recovery), before and after the corrections alike: the discrete treatment of the impedance terms under central difference is the suspect, consistent with the literature's warnings about discretizing absorbing boundary operators explicitly. (iii) A bimaterial benchmark to exercise the tensor impedance and cross-scaling where they should pay off.

7 Summary

- On a *non-overlap* interface, RR Schwarz enforces an elastic condition $\boldsymbol{t} + \alpha\,\boldsymbol{u} = \boldsymbol{g}$; α is a user-supplied stiffness that trades off traction and displacement error.

- Impedance non-overlap Schwarz augments the Robin condition with the Lysmer-Kuhlemeyer absorbing term $Z\,\dot{\boldsymbol{u}}$ using the material's own P-wave impedance $Z = \sqrt{\rho(\lambda + 2\mu)}$. No tuning is required for Z; the displacement-penalty α is optional. Discretely, the only differences from RR are (i) an extra $Z\,c_v$ contribution to the interface tangent and (ii) a partner velocity term in the right-hand side; both are assembled with the same surface operator $\boldsymbol{W}$ and projection $\boldsymbol{P}$ used by RR.

- In the quasi-static limit $c_v \to 0$, impedance non-overlap Schwarz collapses onto RR. For dynamic problems it is non-reflecting at the interface ($|R_{\mathrm{imp}}| < 1$, with $R_{\mathrm{imp}} = 0$ at $\alpha = 0$), whereas RR is fully reflecting ($|R_{\mathrm{RR}}| \equiv 1$). This is the mechanism that removes the reflection-driven instability of classical RR with mixed integrators.

- On an *overlap* boundary, Norma.jl provides two variants: classical DBC coupling, which interpolates the partner displacement (strongly or in L^2) and constrains the overlap DOFs; and impedance overlap, which leaves the overlap DOFs free and enforces the full transmission condition $\boldsymbol{t}_k - \boldsymbol{t}_j + \boldsymbol{Z}(\dot{\boldsymbol{u}}_k - \dot{\boldsymbol{u}}_j) + \alpha(\boldsymbol{u}_k - \boldsymbol{u}_j) = \boldsymbol{0}$, with the partner traction obtained as the variationally consistent boundary flux (20) on node-aligned interfaces (consistently recovered nodal stress otherwise) and the tensor impedance $\boldsymbol{Z}$ built from the partner's P- and S-wave impedances.

- The partner-traction term is what makes the impedance overlap condition consistent with the monodomain solution, and the *quality* of the traction sets the energy error: on the conforming cantilever benchmark the ladder runs 59% (missing $\boldsymbol{t}_j$) $\rightarrow$ 14% (lumped recovery) $\rightarrow$ 1.5% (consistent recovery) $\rightarrow$ 0.08% (consistent flux) over 1 ms, parameter-free (physical impedance, $\alpha = 0$), the final rung matching the 0.1% exact-conservation reference of converged conforming DBC coupling. Converged DBC coupling on *non-conforming* meshes is exponentially unstable (transfer-induced); the impedance variant's velocity-jump dissipation is what keeps it bounded there.